“The Essentials of Computer Organization and Architecture”

Unit 1 – Exercises

1.2.a. How many milliseconds are there in one second?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.b. How many microseconds are there in one second?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.c. How many nanoseconds are there in one millisecond?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.d. How many microseconds are there in one millisecond?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.e. How many nanoseconds are there in one microsecond?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.f. How many kilobytes are there in one gigabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.g. How many kilobytes are there in one megabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.h. How many megabytes are there in one gigabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	µ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.i. How many bytes are there in 20 megabytes?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	µ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.j. How many kilobytes are there in 2 gigabytes?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2*. A RAM Memory of 256 GBytes of capacity has more ____ Bytes than a Hard Disk with 256 GBytes of capacity.

(present only the integer part of the number, without rounding, no use of scientific notation, no numerical powers (2^..., 10^...), no SI symbols (K, M, G, ...); for example, if the value obtained from the calculations is 123456,789, present only 123456).

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
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1.3. System A performs a task in a time of the order of nanoseconds, and system B performs the same task in a time of the order of milliseconds. What is the order of magnitude of the acceleration (speedup) of A relative to B for the same task?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
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Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
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Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
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Note:

If A spends time t_A on a task and B spends time t_B on the same task, the speedup of A relative to B is given by $\mathbf{t_B / t_A}$.

1.7.a. If a transistor currently takes an area of 2 um^2, what area will a transistor of the same type need 2 years from now, according to the 18-month version of Moore's Law? (present the result up to 4 decimal places, truncated)

1.7.a* If an integrated circuit has currently 1 million transistors, how many transistors should it be possible to accommodate in a circuit of the same size, 2 years from now, according to the 18-month version of Moore's Law? (show only the integer part of the result)

1.7.b. What is the influence of the validity of Moore's Law on a programmer's activity?